Some Quantum Computing Heresy

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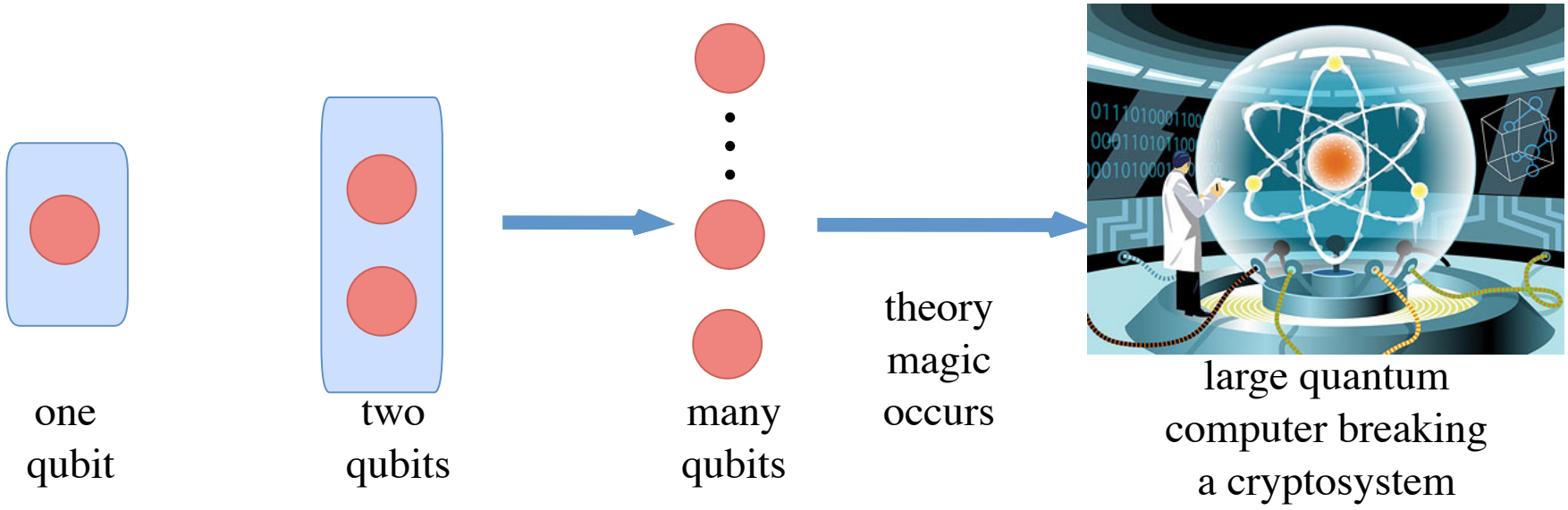
Gregory Crosswhite (Physics), Lukas Svec (Physics)

IARPA visit to UW Nov 16, 2009

The Central Dogma of Quantum Computing

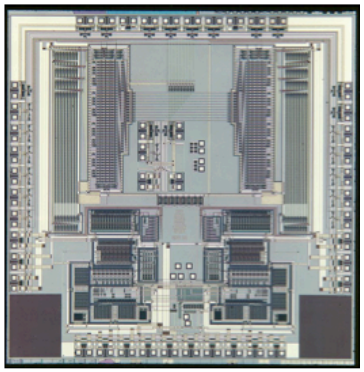
Faulty quantum computing elements can be put together to build a quantum computer with a low probability of failure if the individual elements fail with small enough probability (the threshold)

Therefore we should focus on one and two qubit devices, making all effort to get devices below threshold



Looking Nothing Like the Classical World



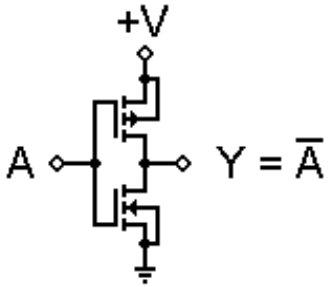


Classical devices need no or little error correction?
No fault-tolerant computing methods?

Observation: The statistical physics of classical computing devices is responsible for their ability to robust store and manipulate data.

A Second Look at the Transistor

Can we build a quantum transistor?



Integrated Circuit Transistors

information propagates spatially across device

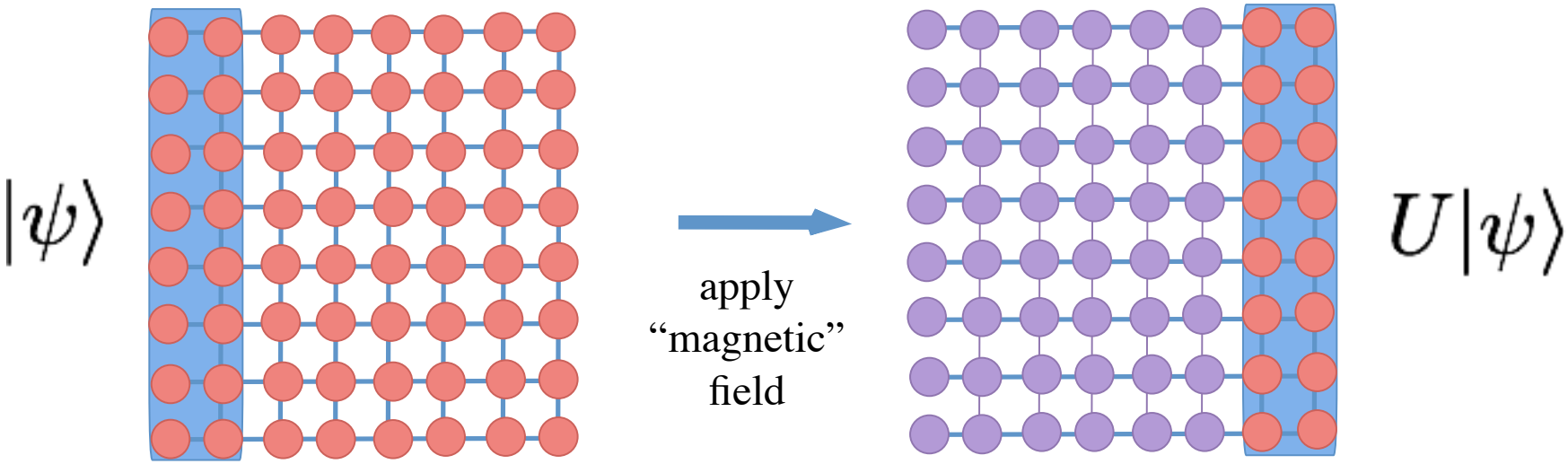
Phase transition used to induce this propagation (on/off = different phases)

Current Quantum Computer Designs

static information carriers, algorithm applied to static carriers (or flying qubits)

"Pulsed" implementations involving precise timing to apply computations

How to Build a Quantum Transistor



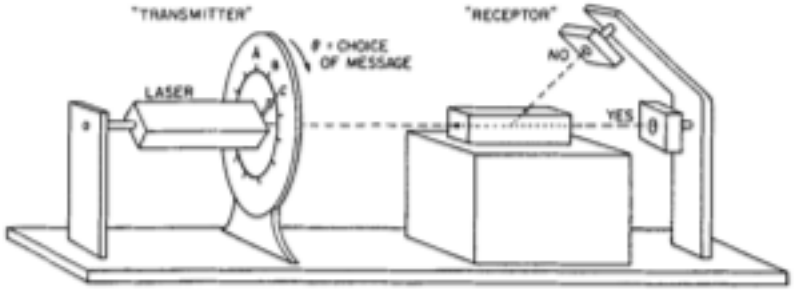
Applied field induces quantum phase transition and quantum information is propagated from input to output with a quantum circuit applied

No pulsing: applied field need only be applied "slowly", errors in timing do not affect result.

Can we build this quantum transistor? D-wave might.

One More Heresy For the Road

The quantum laws of probability can be (nearly) derived from maximizing information in setting where angle is revealed through stochastic “yes”/“no” clicks (Wootters)



Can we invert this observation: derive optimal machine learning algorithms for learning data when it is revealed only through probabilities of outcomes?

Dual motivations for quantum computing